

TZANEEN-WESTFALIA ESTATE, South Africa

WMO: 681880

Lat: 23.7367S

Lon: 30.1127E

Elev: 2940

StdP: 13.20

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	43.5	45.7	25.9	22.2	68.2	30.4	27.4	67.5	10.5	63.9	9.0	63.6	2.2	40	0.255

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.0	93.5	65.8	90.8	66.3	88.0	66.8	73.0	83.2	71.9	82.0	71.0	80.6	4.0	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.1	123.6	77.6	69.1	119.3	76.3	68.2	115.7	75.1	38.8	83.3	37.8	82.2	36.9	80.9	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.5	7.5	5.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	68.4	73.9	73.7	72.3	68.3	64.8	61.3	59.9	63.7	67.8	69.9	72.0	73.4
	DBStd	6.95	3.77	3.72	4.04	4.76	4.06	4.35	4.43	5.39	6.42	6.82	6.32	5.32
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	555	0	0	4	21	52	127	170	89	45	29	14	4
	CDD50	6708	742	663	690	548	458	339	307	426	534	616	660	725
	CDD65	1790	277	244	230	119	45	16	11	50	129	180	225	264
	CDH74	11829	1528	1215	1217	562	267	104	72	444	1139	1609	1807	1865
	CDH80	4020	461	344	339	127	45	11	8	110	419	703	739	714
(14) Wind	WSAvg	2.3	2.2	2.4	2.3	2.2	2.0	2.2	2.2	2.4	2.5	2.6	2.5	2.3
(15) Precipitation	PrecAvg	41.1	7.3	9.1	6.0	1.9	0.7	0.7	0.6	0.4	0.7	2.1	5.0	6.4
	PrecMax	99.1	15.9	46.7	18.5	4.2	4.8	2.7	2.8	1.5	3.6	5.0	10.7	14.1
	PrecMin	20.0	1.6	1.2	2.2	0.2	0.0	0.0	0.1	0.0	0.0	0.2	1.1	2.5
	PrecStd	18.1	4.4	10.4	4.0	1.2	1.1	0.7	0.8	0.4	0.9	1.3	2.5	2.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.2	92.4	90.9	87.2	84.6	81.5	79.9	87.7	94.0	97.0	95.7	96.4
		MCWB	70.8	69.6	67.2	66.5	60.5	55.7	55.8	59.4	60.8	65.1	64.4	69.0
	2%	DB	88.6	87.2	86.9	83.8	80.3	76.8	75.2	83.3	89.2	92.4	93.1	91.1
		MCWB	69.6	70.6	68.1	66.7	61.2	55.7	55.5	59.0	60.8	64.1	65.5	68.8
	5%	DB	85.0	84.3	83.9	80.4	76.8	73.1	71.7	79.1	85.1	87.8	88.8	87.6
		MCWB	69.3	70.0	68.7	65.9	60.9	55.6	55.2	57.8	60.7	64.4	66.6	69.5
	10%	DB	81.8	81.3	80.9	76.9	73.6	69.9	68.2	74.9	80.9	82.9	84.3	83.8
		MCWB	69.1	69.4	68.5	65.4	60.3	55.7	55.0	56.8	60.4	64.2	67.1	69.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.7	73.8	73.3	71.2	65.9	61.7	60.2	65.4	67.2	70.9	72.3	74.0
		MCDB	85.6	84.0	82.6	80.3	75.8	72.2	70.6	75.9	81.1	83.7	86.3	84.5
	2%	WB	72.9	72.2	71.9	69.5	64.3	59.3	58.4	61.9	65.2	68.8	70.4	72.4
		MCDB	82.4	82.0	80.8	78.3	74.0	70.2	67.8	76.9	79.8	81.5	82.0	83.7
	5%	WB	71.4	71.1	70.8	68.1	62.9	57.9	57.1	60.1	63.6	67.3	69.1	71.1
		MCDB	80.5	80.4	79.0	76.5	72.4	67.7	66.4	72.5	77.7	79.8	79.8	82.5
	10%	WB	70.3	70.1	69.6	66.8	61.6	56.6	56.0	58.5	62.1	65.6	67.8	69.9
		MCDB	78.1	78.8	77.1	74.0	70.6	65.6	64.7	69.7	75.3	77.9	78.3	80.9
(35) Mean Daily Temperature Range	5% DB	MDBR	15.0	14.2	15.8	15.4	20.4	21.1	19.9	22.3	21.9	19.9	17.8	16.4
		MCDBR	21.7	20.4	22.3	22.2	25.8	25.9	25.1	29.6	30.5	30.2	27.0	24.5
	5% WB	MCWBR	7.5	7.5	8.6	9.6	10.2	11.1	10.4	10.9	10.5	10.3	9.2	8.2
		MCDBR	16.9	17.2	16.8	18.2	20.5	21.0	19.4	23.0	24.7	22.3	19.7	20.6
		MCWBR	7.0	7.2	7.9	8.4	10.0	10.4	9.6	10.3	10.0	9.3	8.1	7.9
(40) Clear-Sky Solar Irradiance	taub		0.388	0.384	0.375	0.356	0.323	0.308	0.307	0.358	0.416	0.432	0.401	0.397
	taud		2.401	2.413	2.429	2.439	2.484	2.505	2.496	2.340	2.210	2.214	2.338	2.372
	Ebn at Noon		303	300	294	285	283	282	286	280	277	282	297	301
	Edh at Noon		41	39	37	34	29	27	29	36	45	47	43	42
(44) All-Sky Solar Radiation	RadAvg		1973	1901	1707	1468	1383	1233	1310	1535	1746	1840	1907	1936
	RadStd		150	175	147	110	63	84	53	85	113	126	174	167

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page